

Lab 3
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a. Xcelium Simulation Results

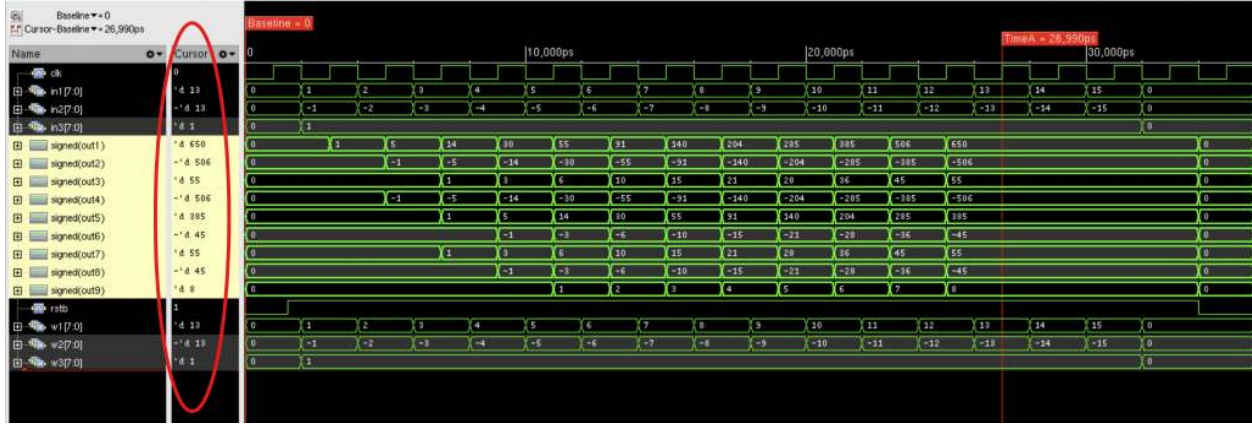


Figure 1: Initial RTL simulation results of Neural Accelerator with no delay annotation

The table below shows the validation table of simulation results. Each cycle is measured on the negative clock edge, where the inputs (IN and W) are set. The result can be seen on the following positive clock edge. In figure 1, we find that on the 13th clock cycle the result has stabilized into their correct values, shown on the left.

Table 1: Validation Table of RTL simulation results

Cycle	Rstb	In	W	Expected Out (1-5)	Expected Out (6-9)	Actual Out (1-5)	Actual Out (6-9)
1	0	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 0 Out2: 0 Out3: 0 Out4: 0 Out5: 0	Out6: 0 Out7: 0 Out8: 0 Out9: 0	Out1: 0 Out2: 0 Out3: 0 Out4: 0 Out5: 0	Out6: 0 Out7: 0 Out8: 0 Out9: 0
2	1	In1: 1 In2: 4 In3: 7	W1: 5 W2: 10 W3: 25	Out1: 5 Out2: -1 Out3: 0 Out4: -1 Out5: 0	Out6: 0 Out7: 0 Out8: 0 Out9: 0	Out1: 5 Out2: -1 Out3: 0 Out4: -1 Out5: 0	Out6: 0 Out7: 0 Out8: 0 Out9: 0
3	1	In1: 2 In2: 5 In3: 8	W1: 10 W2: 20 W3: 30	Out1: 14 Out2: -5 Out3: 1 Out4: -5 Out5: 1	Out6: 0 Out7: 1 Out8: 0 Out9: 0	Out1: 14 Out2: -5 Out3: 1 Out4: -5 Out5: 1	Out6: 0 Out7: 1 Out8: 0 Out9: 0
4	1	In1: 3 In2: 6 In3: 9	W1: 25 W2: 50 W3: 75	Out1: 30 Out2: -14 Out3: 3 Out4: -14 Out5: 5	Out6: -1 Out7: 3 Out8: -1 Out9: 0	Out1: 30 Out2: -14 Out3: 3 Out4: -14 Out5: 5	Out6: -1 Out7: 3 Out8: -1 Out9: 0

5	1	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 55 Out2: -30 Out3: 6 Out4: -30 Out5: 14	Out6: -3 Out7: 6 Out8: -3 Out9: 1	Out1: 55 Out2: -30 Out3: 6 Out4: -30 Out5: 14	Out6: -3 Out7: 6 Out8: -3 Out9: 1
6	1	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 91 Out2: -55 Out3: 10 Out4: -55 Out5: 30	Out6: -6 Out7: 10 Out8: -6 Out9: 2	Out1: 91 Out2: -55 Out3: 10 Out4: -55 Out5: 30	Out6: -6 Out7: 10 Out8: -6 Out9: 2
7	1	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 140 Out2: -91 Out3: 15 Out4: -91 Out5: 55	Out6: -10 Out7: 15 Out8: -10 Out9: 3	Out1: 140 Out2: -91 Out3: 15 Out4: -91 Out5: 55	Out6: -10 Out7: 15 Out8: -10 Out9: 3
8	1	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 204 Out2: -140 Out3: 21 Out4: -140 Out5: 91	Out6: -15 Out7: 21 Out8: -15 Out9: 4	Out1: 204 Out2: -140 Out3: 21 Out4: -140 Out5: 91	Out6: -15 Out7: 21 Out8: -15 Out9: 4
9	1	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 285 Out2: -204 Out3: 28 Out4: -204 Out5: 140	Out6: -21 Out7: 28 Out8: -21 Out9: 5	Out1: 285 Out2: -204 Out3: 28 Out4: -204 Out5: 140	Out6: -21 Out7: 28 Out8: -21 Out9: 5
10	1	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 385 Out2: -285 Out3: 36 Out4: -285 Out5: 204	Out6: -28 Out7: 36 Out8: -28 Out9: 6	Out1: 385 Out2: -285 Out3: 36 Out4: -285 Out5: 204	Out6: -28 Out7: 36 Out8: -28 Out9: 6
11	1	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 506 Out2: -385 Out3: 45 Out4: -385 Out5: 285	Out6: -36 Out7: 45 Out8: -36 Out9: 7	Out1: 506 Out2: -385 Out3: 45 Out4: -385 Out5: 285	Out6: -36 Out7: 45 Out8: -36 Out9: 7
12	1	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 650 Out2: -506 Out3: 55 Out4: -506 Out5: 385	Out6: -45 Out7: 55 Out8: -45 Out9: 8	Out1: 650 Out2: -506 Out3: 55 Out4: -506 Out5: 385	Out6: -45 Out7: 55 Out8: -45 Out9: 8
13	1	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 650 Out2: -506 Out3: 55 Out4: -506 Out5: 385	Out6: -45 Out7: 55 Out8: -45 Out9: 8	Out1: 650 Out2: -506 Out3: 55 Out4: -506 Out5: 385	Out6: -45 Out7: 55 Out8: -45 Out9: 8

14	1	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 650 Out2: -506 Out3: 55 Out4: -506 Out5: 385	Out6: -45 Out7: 55 Out8: -45 Out9: 8	Out1: 650 Out2: -506 Out3: 55 Out4: -506 Out5: 385	Out6: -45 Out7: 55 Out8: -45 Out9: 8
15	1	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 650 Out2: -506 Out3: 55 Out4: -506 Out5: 385	Out6: -45 Out7: 55 Out8: -45 Out9: 8	Out1: 650 Out2: -506 Out3: 55 Out4: -506 Out5: 385	Out6: -45 Out7: 55 Out8: -45 Out9: 8
16	0	In1: 0 In2: 0 In3: 0	W1: 0 W2: 0 W3: 0	Out1: 0 Out2: 0 Out3: 0 Out4: 0 Out5: 0	Out6: 0 Out7: 0 Out8: 0 Out9: 0	Out1: 0 Out2: 0 Out3: 0 Out4: 0 Out5: 0	Out6: 0 Out7: 0 Out8: 0 Out9: 0

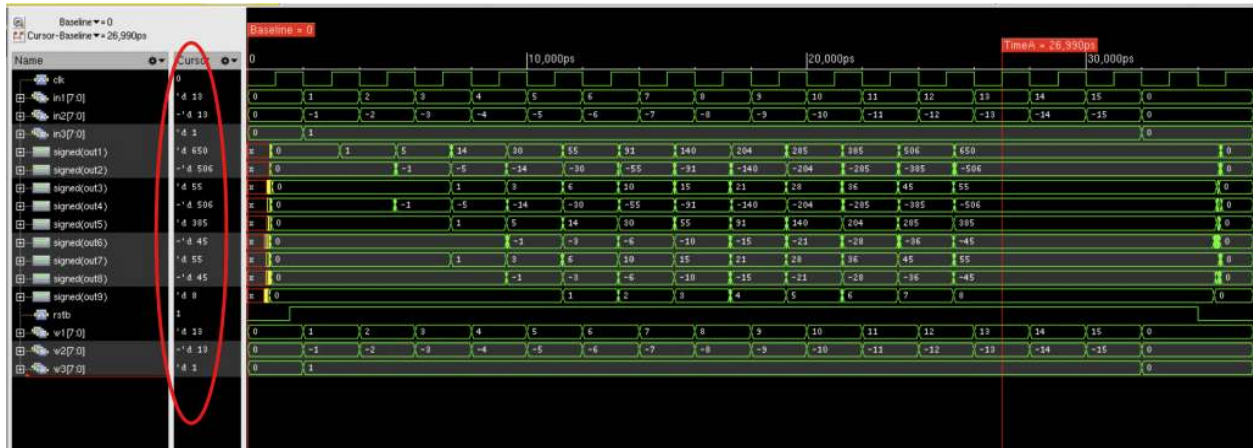


Figure 2: Delay Annotated Simulation of Neural Accelerator module after Backend Design

b. Backend Layout in Innovus

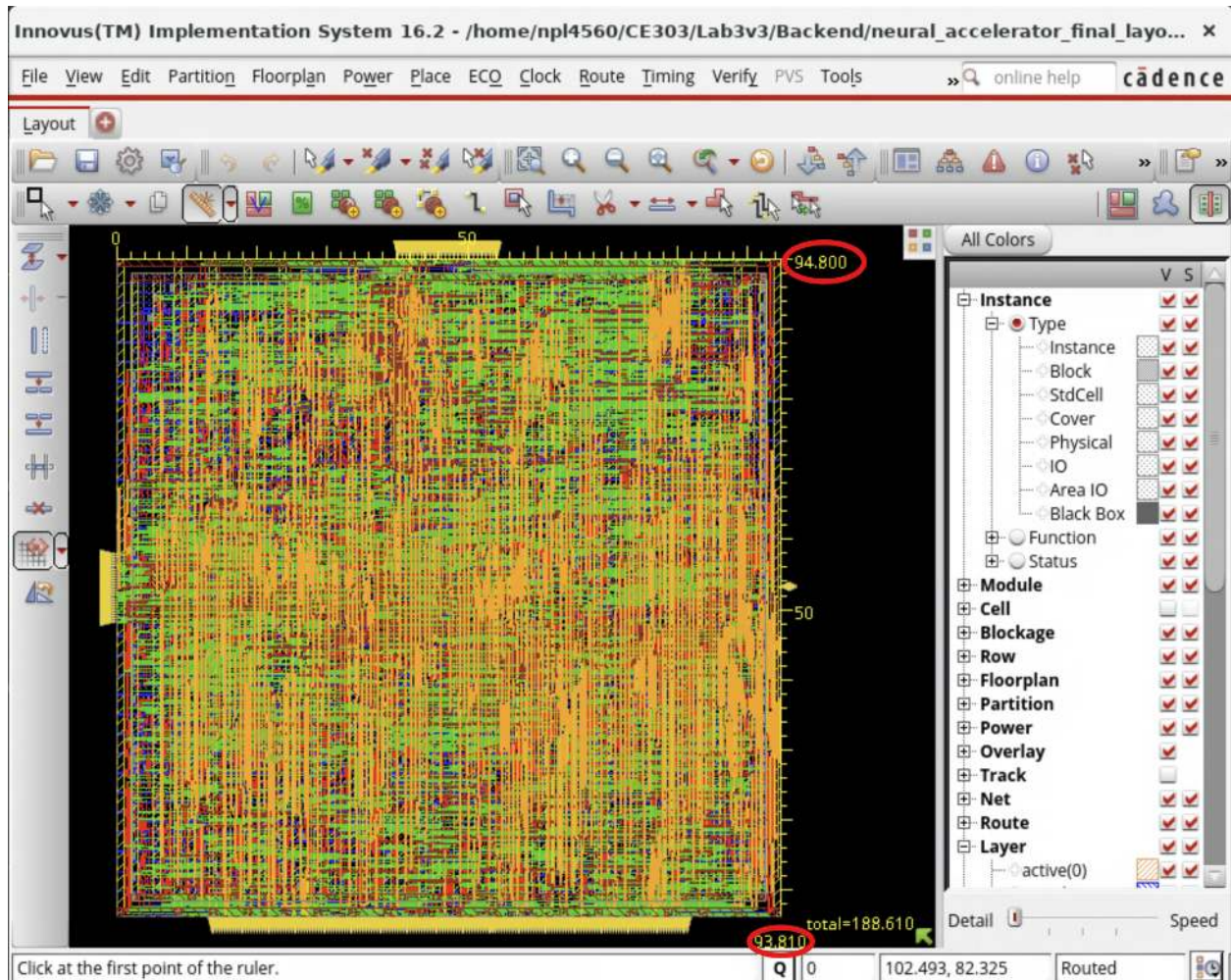


Figure 3: Final Neural Accelerator design, with all placements, buffers, clock and final routing. The x dimension (top right) and y dimension (bottom right) are circled.

c. RTL and testbench codes

RTL file 1: mac_8.v

```
`timescale 1ns/10ps
// Multiplies IN*W and accumulates with internal value
module mac_8(clk, rstb, enc, in, w, out, inR, wR);
    input clk, rstb, enc;
    input signed [7:0] in, w;
    output reg signed [19:0] out;

    // Passing inputs with 1 clock cycle delay
    output reg signed [7:0] inR, wR;
```

```

always @(posedge clk or negedge rstb) begin
    // Handling special cases: rstb and enc (enable calculation)
    if (!rstb) begin
        inR <= 8'b0;
        wR <= 8'b0;
        out <= 20'b0;
    end else begin
        // Update out only if enc is high
        if (enc) begin
            out <= out + (in * w);
        end
        // Pass values regardless of enabled calculation
        inR <= in;
        wR <= w;
    end
end
endmodule

```

RTL file 2: neural_accelerator.v

```
`timescale 1ns/10ps
```

```

module neural_accelerator(in1, in2, in3, w1, w2, w3, clk, rstb, out1, out2, out3, out4, out5, out6,
out7, out8, out9);
    input signed [7:0] in1, in2, in3, w1, w2, w3;
    input clk, rstb;
    output signed [19:0] out1, out2, out3, out4, out5, out6, out7, out8, out9;

    // Internal DFF to count clock cycles (4 bits to count to 13)
    reg [3:0] clk_count;
    // Internal DFF to enable calculation
    reg master_enc;

    // Buffers and wires transferring input delay
    reg signed [7:0] in2B, in3B, in3BB;
    wire signed [7:0] in2B_mac, in3B_in3BB, in3BB_mac;
    reg signed [7:0] w2B, w3B, w3BB;
    wire signed [7:0] w2B_mac, w3B_w3BB, w3BB_mac;

    // Drive all wires with their respective registers
    assign in2B_mac = in2B;

```

```
assign in3B_in3BB = in3B;  
assign in3BB_mac = in3BB;
```

```
assign w2B_mac = w2B;  
assign w3B_w3BB = w3B;  
assign w3BB_mac = w3BB;
```

```
// Wires transferring delayed internal inputs and w  
wire signed [7:0] in1R, in1RR, in2R, in2RR, in3R, in3RR;  
wire signed [7:0] w1R, w1RR, w2R, w2RR, w3R, w3RR;
```

```
// Instantiate 9 MAC modules
```

```
// Row 1 (In1)
```

```
mac_8 mac_00(.clk(clk), .rstb(rstb), .enc(master_enc), .in(in1), .w(w1), .out(out1), .inR(in1R),  
.wR(w1R));  
mac_8 mac_01(.clk(clk), .rstb(rstb), .enc(master_enc), .in(in1R), .w(w2B_mac), .out(out2),  
.inR(in1RR), .wR(w2R));  
mac_8 mac_02(.clk(clk), .rstb(rstb), .enc(master_enc), .in(in1RR), .w(w3BB_mac), .out(out3),  
.inR(), .wR(w3R));
```

```
// Row 2 (In2)
```

```
mac_8 mac_10(.clk(clk), .rstb(rstb), .enc(master_enc), .in(in2B_mac), .w(w1R), .out(out4),  
.inR(in2R), .wR(w1RR));  
mac_8 mac_11(.clk(clk), .rstb(rstb), .enc(master_enc), .in(in2R), .w(w2R), .out(out5),  
.inR(in2RR), .wR(w2RR));  
mac_8 mac_12(.clk(clk), .rstb(rstb), .enc(master_enc), .in(in2RR), .w(w3R), .out(out6), .inR(),  
.wR(w3RR));
```

```
// Row 3 (In3)
```

```
mac_8 mac_20(.clk(clk), .rstb(rstb), .enc(master_enc), .in(in3BB_mac), .w(w1RR), .out(out7),  
.inR(in3R), .wR());  
mac_8 mac_21(.clk(clk), .rstb(rstb), .enc(master_enc), .in(in3R), .w(w2RR), .out(out8),  
.inR(in3RR), .wR());  
mac_8 mac_22(.clk(clk), .rstb(rstb), .enc(master_enc), .in(in3RR), .w(w3RR), .out(out9),  
.inR(), .wR());
```

```
always @(posedge clk or negedge rstb) begin  
    if(!rstb) begin
```

```

// removes Cycle 1 delay; !rstb ensures no calculations occur in each mac module
master_enc <= 1'b1;
clk_count <= 4'b0;

// Reset all input delay registers
in2B <= 8'b0;
in3B <= 8'b0;
in3BB <= 8'b0;
w2B <= 8'b0;
w3B <= 8'b0;
w3BB <= 8'b0;
end else begin
// Enable all MAC modules for first 13 cycles
if (clk_count < 11) begin
    master_enc <= 1'b1;
    clk_count <= clk_count + 1;
end else begin
    // Otherwise, keep clk_count at 13 and turn off calculations
    master_enc <= 1'b0;
end
// Always propagate all input signals, unless on !rstb
in2B <= in2;
in3B <= in3;
in3BB <= in3B;

w2B <= w2;
w3B <= w3;
w3BB <= w3B;
end
end
endmodule

```

RTL file 3: neural_accelerator_tb.v

```
`timescale 1ns/10ps
```

```

module neural_accelerator_tb();
    reg signed [7:0] in1, in2, in3, w1, w2, w3;
    wire signed [19:0] out1, out2, out3, out4, out5, out6, out7, out8, out9;
    reg rstb, clk;

```

```

// Instantiating neural_accelerator
neural_accelerator accelerator_1(
    .in1(in1),
    .in2(in2),
    .in3(in3),
    .w1(w1),
    .w2(w2),
    .w3(w3),
    .clk(clk),
    .rstb(rstb),
    .out1(out1),
    .out2(out2),
    .out3(out3),
    .out4(out4),
    .out5(out5),
    .out6(out6),
    .out7(out7),
    .out8(out8),
    .out9(out9));

initial begin
    clk = 1'b0;
end

always begin
    #1 clk = ~clk;
end

initial begin
    $sdf_annotate("/home/npl4560/CE303/Lab3v4/Backend/neural_accelerator.sdf",
accelerator_1);
    clk = 1'b0;
    rstb = 1'b0;
    in1 = 8'b0;
    in2 = 8'b0;
    in3 = 8'b0;
    w1 = 8'b0;
    w2 = 8'b0;
    w3 = 8'b0;
    // Wait until first positive edge, then release rstb after 0.5ns

```

```
@(posedge clk)
```

```
#0.5
```

```
rstb = 1'b1;
```

```
// New Inputs: 15 active cycles + 1 idle cycle = 16 cycles total
```

```
@(negedge clk) // Cycle 1
```

```
in1 = 8'd1; in2 = -8'd1; in3 = 8'd1;
```

```
w1 = 8'd1; w2 = -8'd1; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 2
```

```
in1 = 8'd2; in2 = -8'd2; in3 = 8'd1;
```

```
w1 = 8'd2; w2 = -8'd2; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 3
```

```
in1 = 8'd3; in2 = -8'd3; in3 = 8'd1;
```

```
w1 = 8'd3; w2 = -8'd3; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 4
```

```
in1 = 8'd4; in2 = -8'd4; in3 = 8'd1;
```

```
w1 = 8'd4; w2 = -8'd4; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 5
```

```
in1 = 8'd5; in2 = -8'd5; in3 = 8'd1;
```

```
w1 = 8'd5; w2 = -8'd5; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 6
```

```
in1 = 8'd6; in2 = -8'd6; in3 = 8'd1;
```

```
w1 = 8'd6; w2 = -8'd6; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 7
```

```
in1 = 8'd7; in2 = -8'd7; in3 = 8'd1;
```

```
w1 = 8'd7; w2 = -8'd7; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 8
```

```
in1 = 8'd8; in2 = -8'd8; in3 = 8'd1;
```

```
w1 = 8'd8; w2 = -8'd8; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 9
```

```
in1 = 8'd9; in2 = -8'd9; in3 = 8'd1;  
w1 = 8'd9; w2 = -8'd9; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 10  
in1 = 8'd10; in2 = -8'd10; in3 = 8'd1;  
w1 = 8'd10; w2 = -8'd10; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 11  
in1 = 8'd11; in2 = -8'd11; in3 = 8'd1;  
w1 = 8'd11; w2 = -8'd11; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 12  
in1 = 8'd12; in2 = -8'd12; in3 = 8'd1;  
w1 = 8'd12; w2 = -8'd12; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 13  
in1 = 8'd13; in2 = -8'd13; in3 = 8'd1;  
w1 = 8'd13; w2 = -8'd13; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 14  
in1 = 8'd14; in2 = -8'd14; in3 = 8'd1;  
w1 = 8'd14; w2 = -8'd14; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 15  
in1 = 8'd15; in2 = -8'd15; in3 = 8'd1;  
w1 = 8'd15; w2 = -8'd15; w3 = 8'd1;
```

```
@(negedge clk) // Cycle 16 (Idle/Hold data)  
in1 = 8'b0; in2 = 8'b0; in3 = 8'b0;  
w1 = 8'b0; w2 = 8'b0; w3 = 8'b0;
```

```
// Final Reset
```

```
@(negedge clk);
```

```
rstb = 1'b0;
```

```
#2 $finish;
```

```
end
```

```
endmodule
```

d. SDC file

```
create_clock -name clk -period 2 -waveform { 0 1 } [get_ports clk]

# ----- Input constraints -----

set_input_delay -clock clk -max 0.5 [get_ports {in1 in2 in3 w1 w2 w3}]
set_input_delay -clock clk -min -0.2 [get_ports {in1 in2 in3 w1 w2 w3}]

# ----- Output constraints -----

set_output_delay -clock clk -max 0.5 [get_ports {out1 out2 out3 out4 out5 out6 out7 out8 out9}]
set_output_delay -clock clk -min -0.2 [get_ports {out1 out2 out3 out4 out5 out6 out7 out8 out9}]

# set_max_delay 0.4 -from [all_inputs] -to [all_outputs]

# Assume 50fF load capacitances everywhere:
set_load 0.050 [all_outputs]
# Set 10fF maximum capacitance on all inputs
set_max_capacitance 0.010 [all_inputs]

# set clock uncertainty of the system clock (skew and jitter)
set_clock_uncertainty -setup 0.03 [get_clocks clk*]
set_clock_uncertainty -hold 0.06 [get_clocks clk*]

# set maximum transition at output ports
set_max_transition 0.07 [current_design]
```

e. Timing Reports

Table 2: Screenshots of timing reports

Setup Timing Report

```
#####
# Generated by: Cadence Innovus 16.20-p002_1
# OS: Linux x86_64(Host ID moore.wot.ece.northwestern.edu)
# Generated on: Thu Jun 11 21:57:28 2026
# Design: neural_accelerator
# Command: timeDesign -postRoute -pathReports -drvReports -slackReports -numPaths 400 -prefix neural_accelerator_postRoute -outDir timingReports
```

```
Path 1: MET Setup Check with Pin mac_00_out_reg[18].CK
Endpoint: mac_00_out_reg[18].SI (*) checked with leading edge of 'clk'
Beginpoint: wlf[1] (v) triggered by leading edge of 'clk'
```

```
Beginpoint: willl
Path Groups: {default}
```

Analysis Vis: an	
Other End Arrival Time	0.161
- Setup	0.061
+ Phase Shift	2.000
- Uncertainty	0.030
= Required Time	2.069
- Arrival Time	1.953
= Slack Time	0.116

```

= Stack Time                                0.116
  Clock Rise Edge                            0.000
+ Input Delay                               0.500
= Beginpoint Arrival Time                    0.500

```

Timing Path:

	Pin	Edge	Net	Cell	Delay	Arrival Time	Required Time
w111		v	w111			0.509	0.616
FE_PMC07_w1_1		v	w111	CLNBUF_X1	0.004	0.504	0.620
FE_PMC07_w1_12		v	FE_PMC07_w1_1	CLNBUF_X1	0.080	0.504	0.701
drc_bufs3019A		v	FE_PMC07_w1_1	INW_X1	0.001	0.595	0.701
mac_00_csa_tree_add_21_28_group1_n_8		^	mac_00_csa_tree_add_21_28_group1_n_8	INW_X1	0.054	0.639	0.755
mac_00_csa_tree_add_21_28_group1_q2371_1788A		^	mac_00_csa_tree_add_21_28_group1_n_8	XOR2_X1	0.000	0.639	0.755
mac_00_csa_tree_add_21_28_group1_1788A2		^	mac_00_csa_tree_add_21_28_group1_n_30	XOR2_X1	0.048	0.687	0.803
FE_OF32_mac_00_csa_tree_add_21_28_group1_n_30A		^	mac_00_csa_tree_add_21_28_group1_n_30	BUF_X1	0.009	0.687	0.803
FE_OF32_mac_00_csa_tree_add_21_28_group1_n_30A2		^	FE_OF32_mac_00_csa_tree_add_21_28_group1_n_30	BUF_X1	0.072	0.759	0.876
mac_00_csa_tree_add_21_28_group1_q2317_5019A		^	FE_OF32_mac_00_csa_tree_add_21_28_group1_n_30	INW2_X1	0.000	0.769	0.876
mac_00_csa_tree_add_21_28_group1_q2317_5019A2		v	mac_00_csa_tree_add_21_28_group1_n_76	INW2_X1	0.057	0.816	0.933
mac_00_csa_tree_add_21_28_group1_q2276_5953A		^	OA122_X1	0.000	0.817	0.933	
mac_00_csa_tree_add_21_28_group1_q2276_5953A2		^	mac_00_csa_tree_add_21_28_group1_n_112	OA122_X1	0.068	0.885	1.001
mac_00_csa_tree_add_21_28_group1_q2246_5703A		^	mac_00_csa_tree_add_21_28_group1_n_112	EA_X1	0.001	0.885	1.002

mac_00_csa_tree_add_21_28_group1_g2246_5703	C1	^	mac_00_csa_tree_add_21_28_group1_n_112	FA_X1	0.001	0.896	1.002
mac_00_csa_tree_add_21_28_group1_g2246_5703	C1	^	mac_00_csa_tree_add_21_28_group1_n_160	FA_X1	0.013	0.886	1.132
mac_00_csa_tree_add_21_28_group1_g2228_8780	A	v	mac_00_csa_tree_add_21_28_group1_n_160	FA_X1	0.000	0.996	1.112
mac_00_csa_tree_add_21_28_group1_g2228_8780	C	v	mac_00_csa_tree_add_21_28_group1_n_187	FA_X1	0.083	0.079	1.195
mac_00_csa_tree_add_21_28_group1_g2207_7118	C1	^	mac_00_csa_tree_add_21_28_group1_n_187	FA_X1	0.000	0.079	1.195
mac_00_csa_tree_add_21_28_group1_g2207_7118	C	v	mac_00_csa_tree_add_21_28_group1_n_213	FA_X1	0.074	1.153	1.260
mac_00_csa_tree_add_21_28_group1_g2201_1309	C1	^	mac_00_csa_tree_add_21_28_group1_n_213	FA_X1	0.000	1.153	1.269
mac_00_csa_tree_add_21_28_group1_g2201_1309	C	v	mac_00_csa_tree_add_21_28_group1_n_220	FA_X1	0.000	1.226	1.342
mac_00_csa_tree_add_21_28_group1_g2199_9682	C1	^	mac_00_csa_tree_add_21_28_group1_n_220	FA_X1	0.000	1.226	1.342
mac_00_csa_tree_add_21_28_group1_g2199_9682	C	v	mac_00_csa_tree_add_21_28_group1_n_223	FA_X1	0.075	1.301	1.417
mac_00_csa_tree_add_21_28_group1_g2198_4547	C1	^	mac_00_csa_tree_add_21_28_group1_n_223	FA_X1	0.000	1.301	1.417
mac_00_csa_tree_add_21_28_group1_g2198_4547	C	v	mac_00_csa_tree_add_21_28_group1_n_226	FA_X1	0.082	1.383	1.499
mac_00_csa_tree_add_21_28_group1_g2195_4206	A	v	mac_00_csa_tree_add_21_28_group1_n_226	A01721_X1	0.000	1.383	1.499
mac_00_csa_tree_add_21_28_group1_g2195_4206	C	v	mac_00_csa_tree_add_21_28_group1_n_229	A01721_X1	0.000	1.454	1.571
FE_0FC35_mac_00_csa_tree_add_21_28_group1_n_229	C1	^	mac_00_csa_tree_add_21_28_group1_n_229	BUF_X1	0.000	1.454	1.571
FE_0FC35_mac_00_csa_tree_add_21_28_group1_n_229	C	^	FE_0FH35_mac_00_csa_tree_add_21_28_group1_n_229	BUF_X1	0.038	1.493	1.609
mac_00_csa_tree_add_21_28_group1_g2191_5019	A1	^	FE_0FH35_mac_00_csa_tree_add_21_28_group1_n_229	OR2_X1	0.000	1.493	1.609
mac_00_csa_tree_add_21_28_group1_g2191_5019	C	^	mac_00_csa_tree_add_21_28_group1_n_233	OR2_X1	0.031	1.524	1.640
mac_00_csa_tree_add_21_28_group1_g2185_2256	A1	^	mac_00_csa_tree_add_21_28_group1_n_233	OA1721_X1	0.000	1.524	1.640
mac_00_csa_tree_add_21_28_group1_g2185_2256	C	^	mac_00_csa_tree_add_21_28_group1_n_239	OA1721_X1	0.000	1.538	1.655
mac_00_csa_tree_add_21_28_group1_g2178_7118	A1	^	mac_00_csa_tree_add_21_28_group1_n_246	OA1721_X1	0.000	1.568	1.685
mac_00_csa_tree_add_21_28_group1_g2178_7118	C	^	mac_00_csa_tree_add_21_28_group1_n_246	OA1721_X1	0.030	1.568	1.685
mac_00_csa_tree_add_21_28_group1_g2176_2391	A1	^	mac_00_csa_tree_add_21_28_group1_n_248	OA1721_X1	0.025	1.594	1.710
mac_00_csa_tree_add_21_28_group1_g2176_2391	C	^	mac_00_csa_tree_add_21_28_group1_n_248	FA_X1	0.000	1.594	1.710
mac_00_csa_tree_add_21_28_group1_g2175_2900	C1	^	mac_00_csa_tree_add_21_28_group1_n_249	FA_X1	0.000	1.665	1.781
mac_00_csa_tree_add_21_28_group1_g2175_2900	C	^	mac_00_csa_tree_add_21_28_group1_n_251	FA_X1	0.000	1.665	1.781
mac_00_csa_tree_add_21_28_group1_g2174_6877	C1	^	mac_00_csa_tree_add_21_28_group1_n_251	FA_X1	0.076	1.741	1.857
mac_00_csa_tree_add_21_28_group1_g2174_6877	C	^	mac_00_csa_tree_add_21_28_group1_n_251	FA_X1	0.000	1.741	1.857
mac_00_csa_tree_add_21_28_group1_g2173_1309	C1	^	mac_00_n_73	FA_X1	0.157	1.898	2.014
mac_00_csa_tree_add_21_28_group1_g2173_1309	C	^	mac_00_n_73	BUF_X16	0.002	1.899	2.016
FE_PHK233_mac_00_n_73	A	^	FE_PHK233_mac_00_n_73	BUF_X16	0.026	1.902	2.042
FE_PHK300_mac_00_n_73	A	^	FE_PHM330_mac_00_n_73	CLKBUF_X1	0.001	1.927	2.043
FE_PHK300_mac_00_n_73	C	^	FE_PHM300_mac_00_n_73	CLKBUF_X1	0.026	1.953	2.069
mac_00_out_reg181_N1	SI	^	FE_PHI300_mac_00_n_73	SDOFFR_X1	0.000	1.953	2.069

```

+-----+
Clock Rise Edge                                0.000
- Beginpoint Arrival Time                      0.000
Other End Path:

```

Pin	Edge	Net	Cell	Delay	Arrival Time	Required Time
clk	~	clk			0.000	-0.116
CTS_ccl_BUF_clk_60_L1	~	clk	CLKBUF_X1	0.000	0.000	-0.116
CTS_ccl_BUF_clk_60_L1_2	~	CTS_66	CLKBUF_X1	0.042	0.042	-0.074
CTS_ccl_BUF_clk_60_L2_2a	~	CTS_66	CLKBUF_X3	0.000	0.042	-0.074
CTS_ccl_BUF_clk_60_L2_2b	~	CTS_65	CLKBUF_X3	0.059	0.097	-0.019
CTS_ccl_BUF_clk_60_L3_2a	~	CTS_55	CLKBUF_X2	0.003	0.009	-0.037
CTS_ccl_BUF_clk_60_L3_2b	~	CTS_58	CLKBUF_X3	0.059	0.158	0.042
mac_00_out_reg181_OK	~	CTS_58	SDFFR_X1	0.003	0.161	0.044

Hold Timing Report

```

#####
# Generated by: Cadence Innovus 16.20-p002.1
# OS: Linux x86_64(Host ID moore.wot.oe.northwestern.edu)
# Generated on: Thu Jun 11 21:57:37 2026
# Design: neural_accelerator
# Command: timeDesign -postRoute -hold -pathReports -slackReports -numPaths 400 -prefix neural_accelerator_postRoute -outDir timingReports
#####
Path 1: MET Early External Delay Assertion
Endpoint: out3[6] (v) checked with leading edge of 'clk'
Beginpoint: mac_02_out_reg[6]_Q0 (v) triggered by leading edge of 'clk'
Path Groups: (default)
Analysis View: an
Other End Arrival Time 0.000
- External Delay -0.200
+ Phase Shift 0.000
+ Uncertainty 0.060
+ Required Time 0.260
Arrival Time 0.261
Slack Time 0.001
Clock Rise Edge 0.000
= Beginpoint Arrival Time 0.000

```

Pin	Edge	Net	Cell	Delay	Arrival Time	Required Time
clk	~	clk			0.000	-0.001
CTS_ccl_BUF_clk_G0_L1_1A	~	clk	CLKBUF_X1	0.000	0.000	-0.001
CTS_ccl_BUF_clk_G0_L1_1Z	~	CTS_66	CLKBUF_X1	0.042	0.042	0.041
CTS_ccl_BUF_clk_G0_L2_1A	~	CTS_66	CLKBUF_X2	0.000	0.042	0.041
CTS_ccl_BUF_clk_G0_L2_1Z	~	CTS_57	CLKBUF_X2	0.054	0.096	0.095
CTS_ccl_BUF_clk_G0_L3_1A	~	CTS_57	CLKBUF_X3	0.001	0.096	0.096
CTS_ccl_BUF_clk_G0_L3_1Z	~	CTS_55	CLKBUF_X3	0.073	0.169	0.169
mac_02_out_reg[6]_CK	~	CTS_55	SDFFR_X1	0.001	0.170	0.169
mac_02_out_reg[6]_Q0	v	out3[6]	SDFFR_X1	0.090	0.260	0.259
out3[6]	v	out3[6]	neural_accelerator	0.001	0.261	0.260

f. Area Report

```

innovus 1> report_area
Depth Name #Inst Area (um^2)
-----
0 neural_accelerator 2896 6146.728

```

Figure 4: Area report of neural_accelerator

g. Verify DRC

```

innovus 69> verify_drc -report neural_accelerator.drc.rpt -limit 1000
#-report neural_accelerator.drc.rpt # string, default="", user setting
*** Starting Verify DRC (MEM: 1470.8) ***

VERIFY DRC ..... Starting Verification
VERIFY DRC ..... Initializing
VERIFY DRC ..... Deleting Existing Violations
VERIFY DRC ..... Creating Sub-Areas
VERIFY DRC ..... Using new threading
VERIFY DRC ..... Sub-Area : 1 of 4
VERIFY DRC ..... Sub-Area : 1 complete 0 Viols.
VERIFY DRC ..... Sub-Area : 2 of 4
VERIFY DRC ..... Sub-Area : 2 complete 0 Viols.
VERIFY DRC ..... Sub-Area : 3 of 4
VERIFY DRC ..... Sub-Area : 3 complete 0 Viols.
VERIFY DRC ..... Sub-Area : 4 of 4
VERIFY DRC ..... Sub-Area : 4 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:00.7 ELAPSED TIME: 1.00 MEM: 26.2M) ***

```

Figure 5: Screenshot of DRC Report with no violations

h. Verify Connectivity

```
innovus 70> verifyConnectivity -type all -error 1000 -warning 50
VERIFY_CONNECTIVITY use new engine.

***** Start: VERIFY CONNECTIVITY *****
Start Time: Mon Jun  1 13:46:06 2026

Design Name: neural_accelerator
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (94.6200, 93.8000)
Error Limit = 1000; Warning Limit = 50
Check all nets

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Mon Jun  1 13:46:06 2026
Time Elapsed: 0:00:00.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols.  0 Wrngs.
(CPU Time: 0:00:00.2  MEM: 0.000M)
```

Figure 6: Screenshot of Connectivity Report with no violations

i. Power Analysis

```
Total Power
-----
Total Internal Power:      5.04720803      60.7513%
Total Switching Power:    3.14252993      37.8254%
Total Leakage Power:      0.11824051       1.4232%
Total Power:              8.30797845
-----
Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:00,
mem(process/total)=851.16MB/851.16MB)
```

j. Path to folder

/home/npl4560/CE303/Lab3v4